

Anthony Ibarra | Electrical & Computer Engineer

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Summary

Electrical and Computer Engineering graduate looking for the opportunity to become a professional in my respective career. Showcasing a strong focus in Embedded Systems, Control theory, Robotics, including pattern recognition involving AI CNN machine learning. And power electronics making me a flexible Embedded Software Engineer with a strong foundation. I possess various knowledge in different fields providing fast adaptable capability within the work environment of a multidisciplinary team. With the eagerness to learn on and off the job, demonstrating reliability and trustworthiness to depend on one another.

EDUCATION

University of Illinois Chicago (UIC)

Jan 2023 - May 2025

Master of Science in Electrical and Computer Engineering

Relevant Coursework: Audio Signal Processing, Intro to Filter Synthesis, Adaptive Digital Filters, Linear Systems Theory & Design, Advanced Computer Communication, Convex Optimization, Mechatronics Embedded Design

University of Illinois Chicago (UIC)

Aug 2017 - May 2022

Bachelor of Science in Computer Engineering

Relevant Coursework: Artificial Intelligence I, Principles of Modern Control & Principles of Auto Control, Pattern Recognition I, Robotics: Algorithm/Control, Embedded Systems

SKILLS

Programming: Python, C/C++, Embedded C, Verilog, RTOS, VHDL, Assembly, ROS2

Software Knowledge: Quartus, MATLAB, simulink, Solidworks, LTSpice, Spice simulation, code-composer, Arduino IDE, VNC viewer, Altium, STM32 CubeMX

Speech: B1 Proficient in spanish

Academic Projects

Research Project, Acoustic & Audio Signal Processing

Jan 2025 - May 2025

Implement a solution for motorcycle riders with some degree of hearing impairment. The starting goal is to remove the dampening from the helmet, providing clear sound while wearing the helmet without adding any unneeded noise or unwanted sounds. In turn this can potentially amplify user awareness while riding or in largely populated areas. Report being in the scope and style of a Signal Processing Society conference paper.

- Develop a new method or solution with concepts to be applied to an existing problem. Perform real world testing and validation of project and research implementation
- Test and use skills and algorithms from acoustic & audio signal processing to achieve the set goal

Self Driving Car, Mechatronics Embedded Design

Jan 2023 - May 2023

- Lead the team and manage all time constraint tasks, development and design tasks, make timely decisions for the team's success in completing the product and hardware development
- Design a motor driver circuit through FET Driver implementation, either of (single fet, half bridge, or full H-bridge), controlled via a PWM input signal from the microcontroller
- Design a Boost Converter for the implementation multiple power systems
- Product testing using Oscilloscope, multimeters, benchtop powersupplies; and Soldering fixes when necessary
- Develop and tune controllers for both the steering and velocity controls (Servo motor, DC motor) respectively
- Design, develop the PCB through Altium Designer and get it manufactured, providing all necessary files such as BOM, gerber files, etc.
- Solder all through hole and SMT components including masking for PCB and Prototype; create any needed jumper wires for connections through crimping and soldering
- Prototype implementation of a perf board circuit, also used as an emergency backup board for PCB failure ensuring continuation of product development
- Implement Sensors and Encoders a filter for the line Camera or velocity measured input through embedded software development and test through debugging microcontroller software

Automated Watering System, Senior Design

August 2021 - May 2022

- Work in a four-student team to prototype a product implementing a automated watering systems plants by taking moisture levels, outdoor weather conditions, and plant information data into account
- Manage team through verbal and written communication to make sure all assignments and goals are completed on time, and submit weekly reports based on progress of project development
- Embedded programming for a Arduino Nano iot 33 to consider watering decisions per plants moisture levels and water consumption
- Create mobile APP (Kivy framework) to display necessary information regarding the system, plants moisture levels and weather conditions
- Create a UDP client-server connection for communication between powered product and the mobile application; providing manual control over a wireless connection
- Implement wireless communication through Bluetooth Protocol (BLE) for consumer and their device

Mobile Robot Control, Robotics: Algorithm/Control

Jan 2022 - May 2022

- Embedded software and firmware development for the robot task completion
- Design kinematics for robot control and awareness for motion planning
- Implement video monitoring through onboard actuated camera
- Path awareness through implementation of arrays of infrared sensors; including robot localization through Aruco markers located around the environment
- Accomplish specific tasks within a constrained location